scientific reports

OPEN

A swarm intelligence-driven hybrid framework for brain tumor classification with enhanced deep features

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Accurate automated classification of brain tumors from magnetic resonance imaging (MRI) is essential for early diagnosis and treatment. This study presents a hybrid framework combining Convolutional Neural Network (CNN) deep features, Large Margin Nearest Neighbor (LMNN) metric learning, and swarm-intelligence optimization for robust four-class classification. Five pretrained CNNs—DenseNet201, MobileNetV2, ResNet50, ResNet101, and InceptionV3—were evaluated on a dataset of 7,023 images categorized as glioma, meningioma, pituitary, healthy. Among these, DenseNet201 provided the highest baseline performance with 92.66% accuracy. LMNN improved feature separability, while Particle Swarm Optimization (PSO) and Grey Wolf Optimizer (GWO) selected compact subsets. The selected features were classified using k-Nearest Neighbor (KNN), Support Vector Machine (SVM), Artificial Neural Network (ANN), and Random Forest (RF). The DenseNet201–LMNN–GWO–KNN configuration, termed DenseWolf-K, achieved the best performance with 99.64% accuracy, establishing it as the optimal implementation of the framework. Robustness and generalizability were further confirmed using an independent external dataset. Model explainability was ensured through feature-level ranking of GWO-selected features and occlusion sensitivity maps, an Explainable Artifical Intelligence (XAI) method. Overall, the proposed DenseWolf-K framework delivers high accuracy, low false-negative rates, compact representation, and enhanced interpretability, representing a reliable and efficient solution for MRI-based brain tumor classification.

Keywords Brain tumor classification, Convolutional neural networks, Metric learning, Swarm intelligence, Feature selection

Brain tumors are among the most critical neurological disorders, leading to significant morbidity and mortality across all age groups[1,2]. Accurate and early diagnosis of these tumors is crucial for effective treatment planning, prognosis assessment, and improving patient outcomes[3]. Magnetic Resonance Imaging (MRI) is the most widely used non-invasive modality for visualizing brain structures due to its high resolution and soft tissue contrast. However, manual interpretation of brain MRI images is time-consuming, subject to inter-observer variability, and highly dependent on radiological expertise[4,5].

Recent advances in artificial intelligence (AI), particularly deep learning (DL) and machine learning (ML), have shown strong potential for automating medical image analysis[6–8]. Convolutional Neural Networks (CNNs) are especially effective in brain tumor classification, as they automatically extract discriminative features from raw imaging data[9–11]. Yet, deep CNN features are often high-dimensional and redundant, leading to increased computational burden and potential overfitting[12–14].

To address these challenges, this study introduces a hybrid framework that integrates CNN-based deep feature extraction, Large Margin Nearest Neighbor (LMNN) metric learning, and swarm intelligence-based feature selection. Five pretrained CNNs—DenseNet201, MobileNetV2, ResNet50, ResNet101, and InceptionV3—were evaluated on a publicly available dataset of 7023 T1-weighted axial MRI images across four classes: glioma, meningioma, pituitary, and healthy. Figure 1 presents representative MRI samples for each category.

CNNs were employed not only for direct classification but also as feature extractors. LMNN was then applied to improve feature separability by bringing same-class samples closer and pushing apart different-class samples[15]. To assess the impact of metric learning on feature selection and classification, two swarm intelligence algorithms—Particle Swarm Optimization (PSO) and Grey Wolf Optimizer (GWO)—were applied both before

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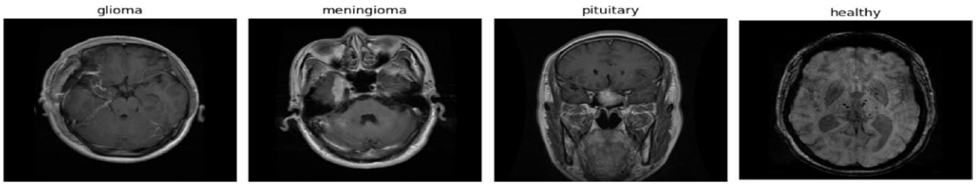


Fig. 1. Representative MRI images from each class in the brain tumor dataset.

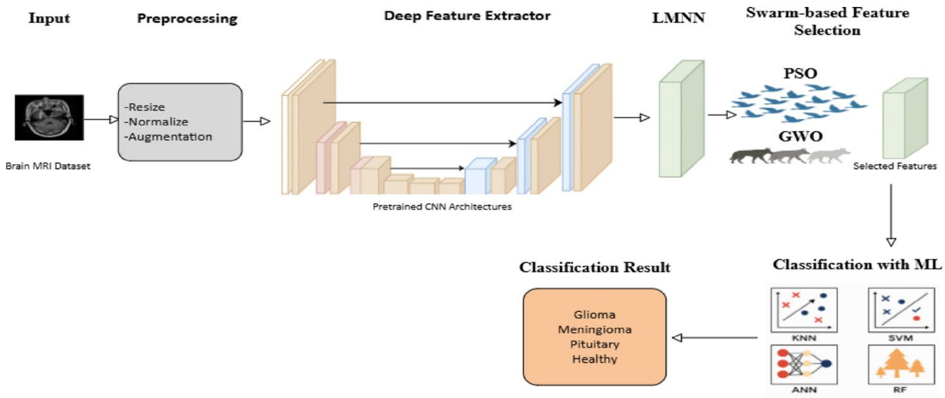


Fig. 2. Flowchart of the present study.

and after LMNN transformation. This dual-phase design constitutes a novel methodological contribution, enabling a rigorous assessment of how metric learning influences optimization-based selection and downstream classification.

The selected subsets were classified using k-Nearest Neighbor (KNN), Support Vector Machine (SVM), Artificial Neural Network (ANN), and Random Forest (RF). The best performance was achieved when GWO was applied to LMNN-transformed DenseNet201 features, reaching 99.64% accuracy with 945 selected features. This highlights the synergy of CNN features, LMNN-driven metric learning, and swarm-based optimization—a pipeline not previously explored in brain tumor classification.

The key contributions of this study are:

- A systematic evaluation of five pretrained CNN architectures to establish classification baselines,
- Application of PSO and GWO for feature selection, enabling robust comparison of optimization strategies,
- A two-stage feature selection pipeline applied before and after LMNN transformation, quantifying the role of metric learning,
- Rigorous evaluation with four machine learning classifiers to ensure generalizability,
- Validation on a large, annotated dataset (7,023 images) and an external dataset, confirming robustness,
- Feature-level and image-level transparency via GWO feature ranking and occlusion sensitivity heatmaps.

The motivation behind this study lies in addressing the urgent clinical need for fast, reliable, and interpretable brain tumor classification tools that minimize false negatives and reduce the workload of radiologists. By integrating methodological innovations with practical applicability, the proposed framework contributes not only to academic progress but also to improved diagnostic support in real-world healthcare environments.

Overall, this study provides a novel and empirically validated hybrid methodology that uniquely integrates deep learning, metric learning, and swarm intelligence algorithms in a unified framework for medical image analysis. The proposed solution is not only highly accurate and computationally efficient but also flexible enough to be extended to other diagnostic imaging applications, thereby contributing meaningfully to the advancement of AI-driven clinical decision-support systems. Figure 2 shows a flowchart of this study.

Related work

Recent advancements in artificial intelligence have significantly accelerated the development of computer-aided diagnostic (CAD) systems for brain tumor classification. Deep CNNs have been widely adopted for the automated analysis of brain MRI images. Pereira et al.[16], for instance, developed a CNN-based segmentation model that achieved competitive results in the BRATS challenge. Ayadi et al.[17] applied deep CNNs for classification and reported accuracy above 94% across multiple datasets. Transfer learning approaches using pretrained models such as VGG16, ResNet50, and InceptionV3 have also shown success[18,19].

Beyond direct classification, CNNs are often used as feature extractors combined with traditional classifiers such as SVM, KNN, and RF[20]. However, high-dimensional feature spaces necessitate feature selection to reduce redundancy and computational cost. Metaheuristic algorithms such as Genetic Algorithms (GA), GWO, and PSO have been applied in this context[21–24].

Metric learning, particularly LMNN, has been explored in image retrieval and face recognition but rarely in medical image classification[15]. To our knowledge, no prior study has simultaneously integrated CNN-based deep features, LMNN metric learning, and swarm intelligence-based feature selection into a unified pipeline. This study addresses that gap by applying PSO and GWO both before and after LMNN transformation, allowing a systematic assessment of metric learning's impact on feature optimization and classification.

Moreover, the selected feature subsets were rigorously evaluated using four machine learning classifiers—KNN, SVM, ANN, and RF—enabling a robust comparison across diverse learning paradigms. This integration of deep learning, metric learning, metaheuristic optimization, and conventional classifiers represents a novel and holistic approach to brain tumor classification, contributing to the development of accurate and interpretable clinical decision-support systems.

Table 1 summarizes recent studies on brain tumor classification using deep learning, feature selection, and metric learning, highlighting the unique contributions of this work.

While prior studies achieved high accuracies with CNN-based models, they often relied on high-dimensional and redundant features, increasing computational complexity and reducing robustness. Most were limited to small datasets or binary classification tasks, leaving multi-class classification underexplored. Crucially, external validation on independent datasets was rarely performed, raising concerns about generalizability. Although optimization algorithms and metric learning techniques such as LMNN have been applied in other domains, their integration in medical imaging—and specifically brain tumor classification—remains largely unexplored.

To bridge these gaps, this study proposes a unified framework combining CNN-based deep feature extraction, LMNN-driven metric learning, and swarm intelligence-based feature selection. Comparative analyses against state-of-the-art methods showed superior accuracy and strong robustness on an external dataset. Transparency was ensured at both feature and image levels: feature importance was quantified via GWO-selected features in LMNN space, while image-level interpretability was provided by occlusion sensitivity analysis, generating class-relevant heatmaps.

The remainder of this paper is organized as follows. "Materials and methods" section describes the datasets, preprocessing, CNN backbones, and hybrid pipeline, including PSO/GWO and LMNN configurations, classifiers, metrics, and cross-validation. "Experimental results and discussions" section presents experimental setup, ablation studies, classification results, comparative analyses, and external validation. "Explainability Analysis" section develops the explainability analysis, and "Conclusion" section concludes with future research directions.

Materials and methods

This section outlines the conceptual foundation of the proposed hybrid framework for brain tumor classification. It introduces the characteristics of the brain MRI dataset and the preprocessing procedures applied to ensure input consistency. Subsequently, the use of pretrained CNNs is explained, highlighting their dual role as both classifiers and deep feature extractors. The methodological components of the study—PSO, GWO, and LMNN—are then detailed, highlighting their role in the dual-phase feature selection strategy. Furthermore, the employed ML classifiers which are KNN, SVM, ANN, and RF, the cross-validation strategy, and the evaluation metrics are described to provide a comprehensive overview of the study design.

References	Dataset	Classes	Methods	Best accuracy
Deepak and Ameer[20]	Fighshare	3	GoogLeNet + KNN	98.0%
Khan et al.[25]	BraTS-2015	2	VGG19	94.06%
Çinar and Yıldırım[18]	Kaggle	2	Hybrid CNN	97.2%
Kang et al. (2021)[26]	Kaggle	2	Pre-Trained CNNs + ML Classifiers	0.9817% with MnasNet + FC
Rasheed et al.[27]	Kaggle	4	CLAHE + Gaussian + CNN; pretrained model comparison	97.85%
Gómez-Guzmán et al.[28]	Kaggle	4	Generic CNN, ResNet50, InceptionV3, InceptionResNetV2, Xception, MobileNetV2, and EfficientNetB0.	97.12% with InceptionV3
Brahim et al.[27]	Kaggle	2	CNN + Adam optimizer	98.0%
Suryawanshi and Patil[29]	Brats 2018 and Sartaj	3	CNN-SVM	98.02% for Brats 96.83% for Sartaj
Reyes and Sánchez[30]	Figshare and Kaggle	3	Custom shallow CNN, VGG, ResNet, EfficientNet, ConvNeXt; Transfer Learning; Fine-Tuning	98.7% with ResNet101 and EfficientNet
Hosny et al.[31]	Fighshare	3	Ensemble model with DenseNet121 and InceptionV3	99.02%
Amoury et al.[32]	Kaggle	4	PSO-optimized CNN	99.19%
Karpakam and Kumareshan[33]	Fighshare	4	DIR-GAN	98.90%

Table 1. Summary of previous studies on brain tumor classification. CLAHE, contrast limited adaptive histogram equalization; DIR-GAN, dual input residual generative adversarial network; FC, fully connected layer.

Dataset and preprocessing

A publicly available brain MRI dataset was used in this study, obtained from Kaggle[34]. The dataset contains 7,023 T1-weighted axial MRI images, categorized into four classes: glioma (1,621 images), meningioma (1,645 images), pituitary (1,757 images), and healthy (2,000 images). The images were collected from multiple patients and annotated by clinical experts. All images were resized to a fixed resolution of 224 × 224 × 3 pixels to ensure compatibility with pretrained CNN architectures. Pixel values were normalized to the [0, 1] range prior to training.

To enhance generalization and mitigate overfitting, data augmentation techniques were applied to the training set, including random rotation (± 10°), horizontal flipping, shear transformation (0.05), zoom range (± 10%), and width/height shift (± 10%).

The dataset was split into training (80%) and testing (20%) subsets using stratified sampling to preserve class distribution. The resulting data served as input for both classification and deep feature extraction stages of the proposed framework.

Pretrained CNN architectures

CNNs are a class of deep learning models that have demonstrated remarkable success in image classification tasks due to their ability to learn spatial hierarchies of features[35]. A typical CNN architecture is composed of the following key layers:

Convolutional layers These layers apply a set of learnable filters to the input image or previous feature maps to extract local features such as edges, textures, and patterns. The filters slide over the input, preserving the spatial relationship between pixels[35].

Activation layers (e.g., ReLU) After each convolution, a non-linear activation function such as the Rectified Linear Unit (ReLU) is applied to introduce non-linearity, which allows the network to learn complex mappings.

Pooling layers Pooling operations (e.g., max pooling) reduce the spatial dimensions of the feature maps, decreasing the number of parameters and computation while retaining the most relevant information.

Batch normalization This layer standardizes the inputs to a layer for each mini-batch, which accelerates training and improves performance by reducing internal covariate shift.

Fully connected (Dense) layers In the final stages, high-level features extracted from previous layers are fed into one or more fully connected layers for classification.

Softmax layer The output layer uses the SoftMax function to produce probability distributions over the predefined classes.

CNNs have the advantage of parameter sharing, sparse connectivity, and translation invariance, making them highly effective for medical image analysis tasks such as brain tumor classification[36].

In this study, five advanced pretrained CNN models were utilized both for direct classification and for feature extraction in the proposed hybrid framework. These models—DenseNet201, MobileNetV2, ResNet50, ResNet101, and InceptionV3—were initialized with ImageNet weights and fine-tuned on the brain MRI dataset. The extracted deep features were subsequently transformed and optimized using a norm-based feature selection method based on the LMNN algorithm.

DenseNet201

DenseNet201 is a deep convolutional neural network that utilizes dense connectivity, meaning each layer is connected to every other layer in a feedforward fashion. This architecture encourages feature reuse, improves gradient flow, and reduces the number of parameters compared to traditional CNNs of similar depth. DenseNet201 consists of 201 layers, organized into dense blocks and transition layers. The model is known for its robustness and high parameter efficiency, making it suitable for complex image analysis tasks[37,38].

MobileNetV2

MobileNetV2 is a lightweight CNN architecture optimized for mobile and embedded vision applications. It is based on depthwise separable convolutions, which decompose standard convolutions into a depthwise and pointwise convolution to reduce computational cost. It also introduces an inverted residual block with linear bottlenecks, enabling the model to maintain representational power while minimizing latency. MobileNetV2 is efficient in terms of both memory usage and inference speed[39].

ResNet50

ResNet50 is a deep residual learning network composed of 50 layers, built using the concept of identity shortcut connections or skip connections. These residual connections allow gradients to propagate more effectively during backpropagation, addressing the vanishing gradient problem that occurs in very deep networks. The architecture includes convolutional blocks with batch normalization and ReLU activation. ResNet50 achieves strong performance in classification tasks with relatively manageable complexity[40].

ResNet101

ResNet101 extends the ResNet architecture to 101 layers, enabling it to model more complex feature hierarchies. It maintains the core principle of residual learning through skip connections, allowing for effective training of very deep networks. ResNet101 offers improved representation learning compared to ResNet50, particularly beneficial for fine-grained image classification problems. It maintains computational efficiency through bottleneck layers and identity mappings[40].

InceptionV3
InceptionV3 is a convolutional neural network architecture developed by Google as an extension of the Inception family. It introduces factorized convolutions, auxiliary classifiers, and label smoothing to improve model efficiency and accuracy. Architecture employs parallel convolutional operations within "Inception modules," enabling the network to capture multi-scale features at each level. InceptionV3 is designed to achieve a balance between high classification performance and computational cost, making it suitable for medical imaging tasks involving complex patterns[41].

In this study, five pretrained CNN architectures—DenseNet201, MobileNetV2, ResNet50, ResNet101, and InceptionV3—were employed for both feature extraction and classification. Each model outputs a distinct number of deep features from its penultimate layer: 1920 features for DenseNet201, 1280 for MobileNetV2, and 2048 for ResNet50, ResNet101, and InceptionV3, respectively. These features were leveraged in the subsequent feature selection and classification phases. The CNN models were fine-tuned under consistent training conditions summarized in Table 2, utilizing the Adam optimizer, a batch size of 32, an initial learning rate of 0.001, and ReLU activations. Training was performed over 10 epochs to balance learning capacity and overfitting risk.

Swarm-based feature selection in CNN and LMNN feature spaces
This section presents the feature selection process based on swarm intelligence techniques, applied to both the original CNN-extracted feature space and the LMNN-transformed metric space. The aim is to identify the most relevant features for brain tumor classification by the leveraging global optimization capabilities of metaheuristic algorithms. The process is divided into four subsections, covering PSO, GWO, the LMNN algorithm, and the overall strategy integrating them.

Particle swarm optimization (PSO)
Particle Swarm Optimization (PSO) is a population-based stochastic optimization technique inspired by the social behavior of bird flocking, originally proposed by Kennedy and Eberhart[42]. In this method, each particle represents a candidate solution, encoded as a binary feature mask vector $x_i \in \{0,1\}^d$, where d is the number of features. The particles move in the search space by updating their velocities and positions as follows:

$$\boldsymbol{v}_i^{t+1} = w \cdot \boldsymbol{v}_i^t + c_1 \cdot r_1 \cdot \left(\boldsymbol{p}_i - \boldsymbol{x}_i^t\right) + c_2 \cdot r_2 \cdot \left(\boldsymbol{g} - \boldsymbol{x}_i^t\right) \tag{1}$$

$$\boldsymbol{x}_i^{t+1} = \boldsymbol{x}_i^t + \boldsymbol{v}_i^{t+1} \tag{2}$$

where $\boldsymbol{v}_i^t$ and $\boldsymbol{x}_i^t$ denote the velocity and position of the i-th particle at iteration t, respectively. The parameter w is the inertia weight that controls the balance between exploration and exploitation, while c_1 and c_2 are the cognitive and social acceleration coefficients. r_1 and r_2 are random values uniformly sampled from $[0,1]$. $\boldsymbol{p}_i$ represents the personal best position of particle i, and $\boldsymbol{g}$ is the global best position identified by the swarm so far. After updating the velocities and positions, the continuous position values are converted into binary feature-selection masks (e.g., using a sigmoid transfer function followed by thresholding), where 1 indicates that a feature is selected and 0 indicates exclusion[43,44].

In this study, PSO algorithm was employed for feature selection, using a swarm size of 30 particles and a maximum of 50 iterations. The inertia weight (w) was set to 0.72, while the cognitive and social acceleration coefficients (c_1 and c_2) were both set to 1.49.

Grey wolf optimizer (GWO)
The Grey Wolf Optimizer (GWO), introduced by Mirjalili et al.[45], is a population-based metaheuristic inspired by the leadership hierarchy and cooperative hunting strategies observed in grey wolf packs. In the algorithm, the population is divided into four groups: alpha (the best solution), beta (the second best), delta (the third best), and omega (the remaining candidates). These top three wolves guide the rest of the search agents toward promising regions in the solution space.

The position of each wolf is updated based on the relative influence of the alpha, beta, and delta wolves, using the following equations:

$$X_1 = X_\alpha - A_1 \cdot |C_1 \cdot X_\alpha - X| \tag{3}$$

$$X_2 = X_\beta - A_2 \cdot |C_2 \cdot X_\beta - X| \tag{4}$$

$$X_3 = X_\gamma - A_3 \cdot |C_3 \cdot X_\gamma - X| \tag{5}$$

Parameter	Value/description
Optimization method	Adam optimizer was used for its adaptive learning rate and computational efficiency.
Number of epochs	Training was performed for 10 epochs to prevent overfitting while allowing adequate convergence.
Batch size	A batch size of 32 was selected to ensure stable gradient updates and efficient memory usage.
Initial learning rate	The learning rate was set to 0.001 to enable smooth and gradual convergence.
Activation function	ReLU activation was applied in all hidden layers to introduce nonlinearity and reduce vanishing gradient issues.

Table 2. CNN training parameters used in this study.

where X denotes the current position vector of a search agent, and X_α, X_β, and X_γ represent the best, second-best, and third-best solutions found so far, respectively. The coefficient vectors A and C are calculated as $A = 2a \cdot r - a$, $C = 2 \cdot r$, where $r \in [0,1]$ is a random vector of the same dimension as X. The control parameter a decreases linearly from 2 to 0 during iterations, allowing a gradual transition from exploration to exploitation[46,47]. The updated position of a search agent is then computed as:

$$X\,(t+1) = \frac{X_1 + X_2 + X_3}{3} \tag{6}$$

In this study, the GWO algorithm was employed for feature selection, using a swarm size of 30 wolves and a maximum of 50 iterations. The algorithm utilizes a linearly decreasing control parameter a, which starts at 2 and gradually reduces to 0 over the course of the optimization process. This dynamic adjustment allows the algorithm to transition from global exploration to local exploitation, effectively guiding the search toward optimal or near-optimal feature subsets.

Large margin nearest neighbor (LMNN)
Large Margin Nearest Neighbor (LMNN) is a supervised metric learning algorithm designed to enhance the performance of the KNN classifier by learning a data-driven distance metric. Specifically, LMNN learns a linear transformation matrix $\boldsymbol{L}$ that maps the original feature space into a new space where samples from the same class are closer together, while samples from different classes are separated by a large margin. The squared Mahalanobis distance between two instances x_i and x_j is computed as:

$$D(x_i,\, x_j) \;=\; ||\, L(x_i \;-\; x_j)\, ||^2 \tag{7}$$

where $L \in \mathbb{R}^{d\times\, d}$ is the transformation matrix and d is the dimensionality of the feature space. This transformation improves class separability, thereby enhancing the effectiveness of subsequent feature selection and classification steps[15,48].

In this study, LMNN was configured with $k = 2$ target neighbors, a learning rate of $1 \times\ 10^-7$, and a maximum of 300 iterations. The convergence tolerance was set to $1 \times\ 10^-9$ to ensure numerical stability and convergence precision during the optimization of the Mahalanobis distance metric. By explicitly enforcing both intra-class compactness and inter-class separation, LMNN provides a more discriminative representation of the feature space.

Swarm-based feature selection strategy
The proposed feature selection strategy consists of a dual-stage process, integrating swarm intelligence algorithms with metric learning to enhance classification accuracy and reduce feature dimensionality. In the first stage, PSO and GWO were independently applied to the high-dimensional feature vectors extracted from various CNN architectures to identify the most informative subset of features. This initial step aimed to optimize the feature space without applying any metric transformation, providing a baseline evaluation of swarm-based selection.

In the second stage, the same CNN features were subjected to a supervised metric learning process using the LMNN algorithm. The LMNN transformation aims to improve class separability by projecting features into a space where intra-class samples are pulled closer and inter-class samples are pushed apart. PSO and GWO were then reapplied to the LMNN-transformed features to explore whether the enhanced geometry of the feature space leads to improved feature selection and classification outcomes.

To ensure a fair comparison between pre- and post-metric learning results, all swarm algorithm hyperparameters—such as swarm size, number of iterations, and convergence criteria—were kept consistent across both stages. The fitness function guiding each optimization process was defined as one minus the classification accuracy of a 1-Nearest Neighbor (1-NN) classifier, encouraging the selection of compact and highly discriminative feature sets. All feature selection masks were encoded as binary vectors, indicating the inclusion or exclusion of individual features.

The step-by-step implementation of the proposed strategy is summarized as follows:
Step 1. Normalize MRI images to the [0, 1] range and resize to 224 × 224 × 3.
Step 2. Apply data augmentation techniques to enhance model generalizability.
Step 3. Split the dataset into training and testing subsets using stratified sampling.
Step 4. For each CNN model (DenseNet201, MobileNetV2, ResNet50, ResNet101, InceptionV3): (a) Load the CNN without the top classifier layer. (b) Add a new classification head and train the model on the training set. (c) Extract deep features from both training and testing sets.
Step 5. Apply PSO and GWO independently to CNN features to perform feature selection.
Step 6. Train and evaluate KNN, SVM, ANN, and RF classifiers using the selected features.
Step 7. Apply LMNN to the training features and transform both training and test features.
Step 8. Reapply PSO and GWO to the LMNN-transformed features.
Step 9. Train and evaluate KNN, SVM, ANN, and RF classifiers again.
Step 10. Record performance metrics: Accuracy, Precision, Recall, Specificity, F1-score, MCC, and Cohen's Kappa.
Step 11. Compare performance before and after LMNN transformation to assess the added value of metric learning.

Swarm intelligence methods (PSO and GWO) were chosen instead of conventional dimensionality reduction approaches (e.g., PCA, LASSO, autoencoders) because they can capture complex non-linear dependencies among features and perform global optimization without assuming linear separability. LMNN was selected as the metric learning component due to its ability to explicitly maximize intra-class compactness and inter-class separability, providing a more discriminative feature space than unsupervised techniques. Furthermore, four different classifiers (KNN, SVM, ANN, RF) were employed to ensure that the evaluation does not depend on a single learning paradigm, but rather reflects robustness across diverse classification strategies. The step-by-step implementation of this dual-stage strategy is summarized in Algorithm 1.

Input: MRI images
Output: Classified tumor type
1. Preprocess images (resize to 224×224×3, normalize, augment).
2. Split dataset into training and testing sets (stratified).
3. For each CNN model (DenseNet201, MobileNetV2, ResNet50, ResNet101, InceptionV3):
   a. Fine-tune CNN on training data.
   b. Extract deep features from training and testing sets.
   c. Apply PSO and GWO for feature selection (Stage 1).
   d. Train classifiers (KNN, SVM, ANN, RF) and record metrics.
   e. Transform features using LMNN.
   f. Reapply PSO and GWO for feature selection (Stage 2).
   g. Train classifiers again and record metrics.
4. Compare results between Stage 1 and Stage 2 to quantify the added value of LMNN.
5. Evaluate final performance using Accuracy, Precision, Recall, Specificity, F1 -score, MCC, and Cohen 's Kappa.

Algorithm 1. Hybrid CNN-LMNN-swarm-based feature selection framework.

Algorithm 1 provides a clear outline of the proposed pipeline, ensuring methodological transparency and reproducibility. This dual-stage design, in which swarm intelligence algorithms are applied both before and after LMNN transformation, represents a novel integration of optimization and metric learning. To the best of my knowledge, no prior studies in brain tumor classification have systematically compared the impact of swarm-based feature selection in both the raw and LMNN-transformed spaces. This explicit pre- and post-metric learning evaluation, coupled with the use of two different swarm algorithms (PSO and GWO) within the same framework, highlights the uniqueness of the proposed approach and constitutes a key methodological contribution of this study.

Classification algorithms

In the final stage of the proposed brain tumor classification framework, several supervised ML algorithms—KNN, SVM, ANN, RF—were utilized to evaluate the discriminative capacity of the selected features. Each classifier brings unique strengths in terms of model complexity, interpretability, and generalization ability, making them suitable for comparative analysis in medical imaging tasks.

K-nearest neighbor (KNN)

K-Nearest Neighbor (KNN) is a non-parametric, instance-based learning algorithm that classifies data points based on the majority class among their nearest neighbors in the feature space. Its strength lies in its simplicity and effectiveness, particularly when applied to well-distributed, low-noise datasets. As KNN does not involve an explicit training phase, it offers computational efficiency during model construction; however, its inference time may increase significantly with large-scale datasets due to the need to compute distances to all training samples[49].

Support vector machine (SVM)

Support Vector Machine (SVM) is a supervised machine learning algorithm widely used for classification and regression tasks. The fundamental idea behind SVM is to find the optimal hyperplane that maximally separates data points from different classes. In the case of linearly separable data, SVM constructs a decision boundary with the largest possible margin, enhancing the model's generalization ability. For non-linear classification problems, SVM employs kernel functions—such as radial basis function (RBF), polynomial, or sigmoid—to implicitly map the input data into a higher-dimensional space where linear separation is feasible[50].

Artificial neural network (ANN)

ANNs are inspired by the structure and function of biological neural networks. They consist of layers of interconnected artificial neurons capable of capturing non-linear relationships in data[51]. In this study, a multi-layer feedforward ANN with ReLU activation was used, followed by a softmax output layer for multiclass classification. ANNs are highly flexible and perform well when trained with optimized features and sufficient data.

Random forest (RF)
Random Forest (RF) is a robust ensemble learning algorithm that constructs multiple decision trees during training and combines their predictions through majority voting to perform classification. By utilizing bootstrap sampling and random feature selection at each node, RF effectively reduces overfitting and improves generalization performance. It is particularly resilient to noise and outliers, and it performs well on high-dimensional datasets. Moreover, its ability to quantify features enhances interpretability, making it especially suitable for clinical and biomedical classification tasks[52].

The hyperparameters used for configuring the KNN, SVM, ANN, and RF classifiers in this study are summarized in Table 3. These parameter settings were selected based on commonly accepted best practices in the literature to ensure fair comparison and optimal performance across models.

Cross-validation strategy
To evaluate model performance and prevent overfitting, a five-fold stratified cross-validation scheme was employed. The dataset was partitioned into five subsets with preserved class distributions. In each iteration, four folds were used for training and one for testing, ensuring that every sample was used for evaluation exactly once. Final performance metrics were computed by averaging results across all folds, enabling fair and robust comparison of different models[53].

Performance metrics
To evaluate the classification performance of the proposed hybrid framework, several widely accepted performance metrics were employed, including Accuracy, Precision, Recall (Sensitivity), Specificity, F1-Score, Matthews Correlation Coefficient (*MCC*), and Cohen's Kappa. These metrics are derived from the elements of the confusion matrix: True Positive (*TP*), False Positive (*FP*), True Negative (*TN*), and False Negative (*FN*)[38,54].

Although these formulas were initially defined for binary classification problems, this study adopts their generalized multi-class versions by implementing a one-vs- strategy. In this approach, each class is individually treated as the positive class, and all remaining classes are grouped as negative. The metrics are calculated separately for each class, and the results are aggregated using macro-averaging, unless otherwise specified.

The mathematical definitions of the employed metrics are presented in Eqs. (8)–(15):

$$Accuracy = p_a = \frac{TP + TN}{TP + TN + FP + FN} \tag{8}$$

$$F1 - score = \frac{2 \times TP}{2 \times TP + FP + FN} \tag{9}$$

$$Sensitivity = \frac{TP}{TP + FN} \tag{10}$$

$$Specificity = \frac{TN}{TN + FP} \tag{11}$$

Classifier	Parameter	Value
KNN	Number of neighbors (k)	1
	Distance metric	Euclidean
	Weight function	Uniform
SVM	Kernel type	RBF
	Regularization parameter (C)	1.0
	Kernel coefficient	Scale
ANN	Hidden layers structure	(128, 64) neurons
	Activation function	ReLU (hidden), Softmax (output)
	Optimizer	Adam
	Learning rate	Default (≈ 0.001)
	Number of epochs	300
	Batch size	32
RF	Number of trees	100
	Maximum depth	None
	Maximum features	Auto (√features)
	Minimum samples split	2

Table 3. Classifier parameters.

$$Precision = \frac{TP}{TP + FP} \tag{12}$$

$$MCC = \frac{(TP \times TN) - (FP \times FN)}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \tag{13}$$

$$P_e = \frac{((TP + TN)(TP + FP) + (FP + TN)(TP + FN))}{(TP + TN + FP + FN)^2} \tag{14}$$

$$kappa = max\left(\frac{p_a - P_e}{1 - P_e}, \frac{P_e - p_a}{1 - p_a}\right) \tag{15}$$

These metrics offer a comprehensive and interpretable framework for evaluating classification performance in multi-class problems. Accuracy quantifies the proportion of correctly classified instances. F1-score represents the harmonic meaning of precision and recall, balancing the trade-off between false positives and false negatives. Sensitivity (Recall) measures the true positive rate, which is particularly important in medical imaging to minimize false negatives (e.g., missing an actual tumor). Specificity reflects the model's ability to correctly identify negatives, thereby reducing unnecessary clinical interventions. Precision evaluates the correctness of positive predictions, ensuring that cases labeled as tumors are truly positive. MCC is a balanced measure that incorporates all four confusion matrix categories and remains reliable in the presence of class imbalance. Cohen's Kappa assesses the agreement between predicted and actual labels while correcting for agreement that may occur by chance.

Collectively, these performance indicators and their formal definitions not only provide a thorough and fair evaluation of the proposed method but also highlight its clinical significance. By reducing false negatives and ensuring robust classification across tumor and healthy categories, these metrics facilitate reliable comparisons with existing state-of-the-art approaches in medical image classification tasks.

Experimental results and discussions

In this study, five pretrained convolutional neural network (CNN) architectures—DenseNet201, MobileNetV2, ResNet50, ResNet101, and InceptionV3—were evaluated for their performance in brain tumor classification using a five-fold cross-validation strategy. To assess the stability of the training process and to detect any risk of overfitting, training and validation accuracy and loss trends were closely monitored, as depicted in Fig. 3. The figure illustrates that for all models, validation accuracy closely follows training accuracy, and validation loss consistently decreases alongside training loss, indicating that the models are learning effectively without signs of overfitting. Performance metrics, including accuracy, sensitivity, specificity, precision, F1-score, MCC, and Cohen's kappa coefficient, were computed for each of the five pretrained CNN architectures. The results are detailed in Table 4, offering a comparative assessment of the classification capabilities of each model.

DenseNet201 achieved the best performance, with 91.60% accuracy, 91.94% precision, 97.21% specificity, MCC of 0.8889, and Cohen's kappa of 0.8876. Although it required the longest training time (2110 s), its superior metrics justify the computational cost in applications where diagnostic precision is critical. MobileNetV2 followed closely, achieving 90.67% accuracy, 90.54% F1-score, and the shortest training time (1755 s). Its high specificity (96.90%) and efficiency make it suitable for resource-constrained environments. ResNet50 (87.68% accuracy, MCC 0.8354) and ResNet101 (84.27% accuracy) showed moderate performance, indicating that deeper residual connections do not guarantee improvements without extensive tuning. InceptionV3 recorded 85.40% accuracy and 95.13% specificity, with an MCC of 0.8067, performing reasonably well but below DenseNet201 and MobileNetV2. Overall, DenseNet201 consistently outperformed the other models, while MobileNetV2 provided a strong balance between accuracy and computational efficiency.

While all CNN models demonstrated promising classification ability, it was observed that not all extracted features equally contributed to model decisions. Therefore, these CNNs were subsequently employed as deep feature extractors in the second phase of the study. To enhance classification performance, a hybrid framework incorporating advanced feature selection and classification methods was proposed.

The classification performance of the proposed hybrid framework was evaluated using various combinations of CNN-based feature extractors, metric learning via LMNN, and swarm intelligence-based feature selection algorithms (PSO and GWO). Classification results were assessed using KNN, SVM, ANN, and RF classifiers, with accuracy, sensitivity, specificity, precision, F1-score, MCC, Kappa, feature count, and computation time recorded for all configurations, as presented in Tables 5, 6, 7, 8 and 9.

According to Table 5, which presents the classification results using DenseNet201-extracted features, it is evident that applying LMNN prior to feature selection yields a significant performance boost. Notably, the combination of LMNN and GWO with the KNN classifier achieved the highest classification accuracy of 99.64%, along with strong scores across other metrics (sensitivity: 0.9964, specificity: 0.9988, MCC: 0.9952). Without LMNN, the same feature selectors produced slightly lower accuracies. GWO selected fewer features pre- and post-LMNN compared to PSO, which also corresponded with shorter processing times. This shows that GWO not only yields higher accuracy in some settings but does so with more compact feature subsets and more efficient computation. PSO, while delivering competitive results, generally required more time and selected larger subsets, particularly post-LMNN.

In Table 6, MobileNetV2-based models yielded lower performance after LMNN transformation, particularly in SVM, ANN, and RF classifiers. For example, LMNN–PSO–SVM dropped to 87.26% accuracy, compared to 92.38% without LMNN. The highest performance with MobileNetV2 was achieved by GWO + KNN (98.93%),

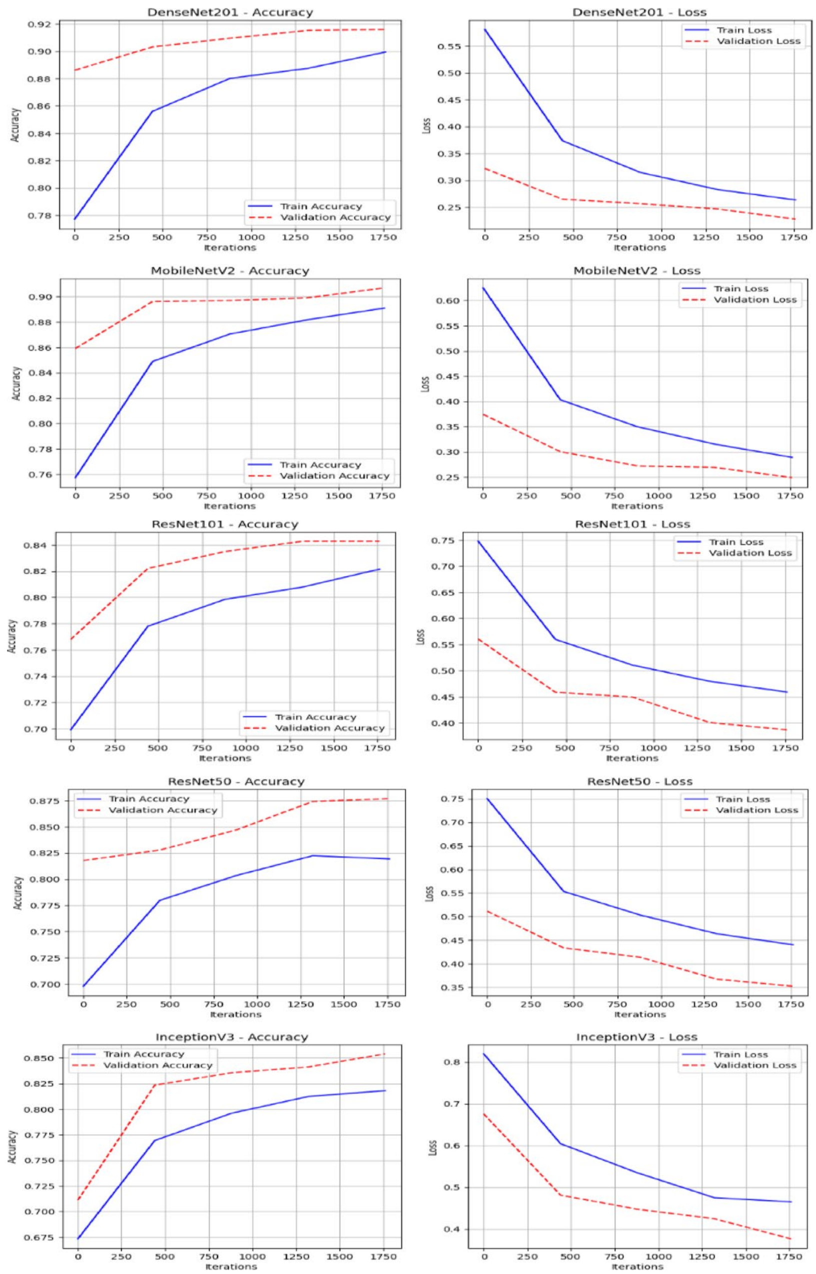


Fig. 3. Accuracy and loss curves for training and validation sets across five pretrained CNN architecture.

Model	Accuracy	Sensitivity	Specificity	Precision	F1-Score	MCC	Kappa	Time (s)
DenseNet201	0.9160	0.9160	0.9721	0.9194	0.9156	0.8889	0.8876	2110
MobileNetV2	0.9067	0.9067	0.9690	0.9067	0.9054	0.8761	0.8752	1755
ResNet50	0.8768	0.8768	0.9591	0.8750	0.8756	0.8354	0.8352	1789
ResNet101	0.8427	0.8427	0.9478	0.8383	0.8393	0.7902	0.7894	1863
InceptionV3	0.8540	0.8540	0.9513	0.8564	0.8514	0.8067	0.8044	1890

Table 4. Classification results of pretrained CNN architectures with 5-fold cross-validation.

Feature selector (Number of selected features)	Classifier	Accuracy	Sensitivity	Specificity	Precision	F1-Score	MCC	Kappa	Time (s)
GWO (532)	KNN	0.9879	0.996	0.9881	0.9881	0.9879	0.9838	0.9839	1292
	SVM	0.9132	0.9714	0.9127	0.9127	0.9129	0.8839	0.8839	1308
	ANN	0.9601	0.9869	0.9605	0.9605	0.9601	0.9467	0.9468	1739
	RF	0.9431	0.9811	0.9448	0.9448	0.9429	0.9238	0.9244	1373
PSO (936)	KNN	0.9879	0.9879	0.996	0.9879	0.9879	0.9838	0.9838	1366
	SVM	0.9359	0.9359	0.9789	0.9367	0.9362	0.9144	0.9145	1384
	ANN	0.9644	0.9644	0.9883	0.9651	0.9645	0.9524	0.9526	1479
	RF	0.9473	0.9473	0.9826	0.9495	0.9474	0.9296	0.9302	1465
LMNN + GWO (836)	KNN	0.9964	0.9964	0.9988	0.9964	0.9964	0.9952	0.9952	7146
	SVM	0.9694	0.9694	0.9899	0.9693	0.9693	0.9591	0.9591	7159
	ANN	0.9765	0.9765	0.9922	0.9767	0.9765	0.9686	0.9687	7175
	RF	0.9431	0.9431	0.9812	0.944	0.943	0.9239	0.9242	7237
LMNN + PSO (945)	KNN	0.9893	0.9893	0.9965	0.9894	0.9893	0.9857	0.9858	6526
	SVM	0.9125	0.9125	0.9712	0.9138	0.913	0.883	0.8831	6538
	ANN	0.9559	0.9559	0.9855	0.9563	0.956	0.941	0.9411	6558
	RF	0.911	0.911	0.9707	0.9106	0.9104	0.881	0.8813	6568

Table 5. Classification results with DenseNet201-based features.

Feature selector (Number of selected features)	Classifier	Accuracy	Sensitivity	Specificity	Precision	F1-Score	MCC	Kappa	Time (s)
GWO (394)	KNN	0.9893	0.9893	0.9965	0.9894	0.9893	0.9857	0.9858	917
	SVM	0.9125	0.9125	0.9712	0.9138	0.9130	0.8830	0.8831	929
	ANN	0.9559	0.9559	0.9855	0.9563	0.9560	0.9410	0.9411	949
	RF	0.9110	0.9110	0.9707	0.9106	0.9104	0.8810	0.8813	959
PSO (630)	KNN	0.9815	0.9815	0.9939	0.9818	0.9816	0.9753	0.9753	1115
	SVM	0.9238	0.9238	0.9750	0.9237	0.9237	0.8982	0.8982	1127
	ANN	0.9516	0.9516	0.9841	0.9518	0.9516	0.9353	0.9353	1415
	RF	0.9146	0.9146	0.9719	0.9148	0.9142	0.8858	0.8861	1173
LMNN + GWO (497)	KNN	0.9765	0.9765	0.9923	0.9767	0.9765	0.9686	0.9687	1669
	SVM	0.8626	0.8626	0.9549	0.8621	0.8618	0.8164	0.8168	1690
	ANN	0.9253	0.9253	0.9753	0.9245	0.9248	0.9001	0.9001	1698
	RF	0.8868	0.8868	0.9627	0.8859	0.8855	0.8486	0.8492	1741
LMNN + PSO (630)	KNN	0.9744	0.9744	0.9916	0.9748	0.9744	0.9658	0.9659	1776
	SVM	0.8726	0.8726	0.9583	0.8746	0.8733	0.8298	0.8300	1790
	ANN	0.9210	0.9210	0.9740	0.9205	0.9205	0.8944	0.8945	1795
	RF	0.8875	0.8875	0.9628	0.8863	0.8862	0.8496	0.8500	1856

Table 6. Classification results with MobileNet-V2 -based features.

underscoring GWO's ability to achieve high performance with minimal dimensionality. This suggests that the LMNN transformation does not universally enhance feature spaces and may negatively impact architectures where initial features are already optimized. Furthermore, LMNN-based configurations incurred greater computational costs (up to 1856 s), while selecting comparable or fewer features than non-LMNN setups, reinforcing that the transformation must be selectively applied.

Table 7 illustrates classification outcomes using ResNet50 features. Here, LMNN–GWO–KNN achieved a modest improvement, reaching 96.65% accuracy. GWO outperformed PSO in terms of classification accuracy in LMNN-transformed space, particularly with KNN and ANN. Interestingly, feature counts and processing time were consistently higher for PSO, reinforcing the notion that GWO offers better compactness-efficiency trade-offs.

Table 8 reports on ResNet101, where overall performance was lower than DenseNet201 but still competitive. LMNN–GWO–ANN reached 92.88% accuracy, showing that ANN benefits more from LMNN in this setup than other classifiers. GWO again demonstrated lower feature counts and computation times compared to PSO, while maintaining or improving accuracy.

In Table 9, InceptionV3 performed best with GWO–KNN (96.87%). However, applying LMNN to InceptionV3 features led to marginal or even negative effects, especially with RF (accuracy declined to 81.85%). This indicates that InceptionV3 features might already exhibit high discriminative power and that further transformation may

Feature selector (Number of selected features)	Classifier	Accuracy	Sensitivity	Specificity	Precision	F1-Score	MCC	Kappa	Time (s)
GWO (719)	KNN	0.9644	0.9644	0.9882	0.9643	0.9642	0.9524	0.9525	1417
	SVM	0.8021	0.8021	0.9348	0.8038	0.8014	0.7358	0.7368	1446
	ANN	0.8157	0.8157	0.9391	0.8221	0.8085	0.7534	0.7593	2477
	RF	0.8875	0.8875	0.9630	0.8865	0.8857	0.8496	0.8504	1447
PSO (1010)	KNN	0.9630	0.9630	0.9878	0.9627	0.9628	0.9505	0.9506	1806
	SVM	0.8142	0.8142	0.9387	0.8129	0.8118	0.7518	0.7529	1841
	ANN	0.8342	0.8342	0.9451	0.8379	0.8326	0.7781	0.7800	2074
	RF	0.8940	0.8940	0.9651	0.8920	0.8918	0.8582	0.8589	1842
LMNN + GWO (665)	KNN	0.9665	0.9665	0.9890	0.9665	0.9665	0.9553	0.9553	7037
	SVM	0.8740	0.8740	0.9585	0.8737	0.8728	0.8316	0.8323	7064
	ANN	0.9288	0.9288	0.9765	0.9289	0.9283	0.9048	0.9052	7188
	RF	0.8925	0.8925	0.9646	0.8908	0.8910	0.8563	0.8567	7102
LMNN + PSO (1021)	KNN	0.9601	0.9601	0.9869	0.9599	0.9600	0.9467	0.9467	7416
	SVM	0.8883	0.8883	0.9632	0.8889	0.8874	0.8507	0.8514	7453
	ANN	0.9253	0.9253	0.9753	0.9251	0.9245	0.9001	0.9005	7524
	RF	0.9032	0.9032	0.9681	0.9021	0.9019	0.8706	0.8710	7499

Table 7. Classification results with ResNet-50 -based features.

Feature selector (Number of selected features)	Classifier	Accuracy	Sensitivity	Specificity	Precision	F1-Score	MCC	Kappa	Time (s)
GWO (638)	KNN	0.9573	0.9573	0.9859	0.9570	0.9571	0.9429	0.9429	1289
	SVM	0.8064	0.8064	0.9360	0.8045	0.8050	0.7412	0.7415	1313
	ANN	0.8527	0.8527	0.9513	0.8514	0.8492	0.8029	0.8047	1553
	RF	0.8890	0.8890	0.9634	0.8877	0.8878	0.8515	0.8518	1311
PSO (1043)	KNN	0.9559	0.9559	0.9854	0.9557	0.9556	0.9410	0.9411	1698
	SVM	0.8192	0.8192	0.9403	0.8170	0.8175	0.7583	0.7587	1735
	ANN	0.8633	0.8633	0.9549	0.8608	0.8612	0.8172	0.8178	2035
	RF	0.8883	0.8883	0.9632	0.8868	0.8873	0.8506	0.8507	1726
LMNN + GWO (611)	KNN	0.9587	0.9587	0.9864	0.9585	0.9585	0.9448	0.9449	5236
	SVM	0.8555	0.8555	0.9525	0.8567	0.8551	0.8070	0.8077	5262
	ANN	0.9267	0.9267	0.9758	0.9263	0.9264	0.9020	0.9021	5369
	RF	0.8890	0.8890	0.9634	0.8873	0.8879	0.8515	0.8517	5294
LMNN + PSO (1061)	KNN	0.9488	0.9488	0.9831	0.9486	0.9486	0.9315	0.9315	5709
	SVM	0.8783	0.8783	0.9599	0.8790	0.8782	0.8374	0.8377	5747
	ANN	0.9288	0.9288	0.9764	0.9285	0.9281	0.9048	0.9052	5859
	RF	0.8918	0.8918	0.9644	0.8904	0.8910	0.8553	0.8554	5790

Table 8. Classification results with ResNet-101-based features.

introduce noise or redundancy. Additionally, GWO-selected feature subsets were consistently smaller, indicating better convergence and selection efficiency.

Figure 4 presents a heatmap of classification accuracy across all configurations, with color intensity encoding the accuracy of each setting. DenseNet201–LMNN–GWO–KNN (DenseWolf-K) clearly stands out as the best-performing model, validating its robustness. MobileNetV2 and ResNet101 displayed high variability across configurations, further supporting the need for model-specific feature refinement.

Figure 5 shows the optimization progress of GWO and PSO in terms of loss reduction. Both algorithms converge rapidly within the first 20–30 iterations, with GWO demonstrating a slightly smoother and more stable optimization trajectory. These curves emphasize the effectiveness of both optimization methods, though GWO offers better convergence consistency.

Figure 6 illustrates the confusion matrix for the best-performing pipeline DenseWolf-K. It shows near-perfect classification with only four misclassifications across all test samples, underlining the discriminative capability and reliability of the selected features.

To validate its effectiveness, the proposed framework was benchmarked against recent state-of-the-art methods reported in the literature (Table 10).

As shown in Table 10, traditional deep learning models such as Generic CNN[28] or InceptionV3[28] provide strong baselines with accuracies of 81.05% and 97.12%, respectively. More advanced hybrid models, e.g.,

Feature selector (Number of selected features)	Classifier	Accuracy	Sensitivity	Specificity	Precision	F1-Score	MCC	Kappa	Time (s)
GWO (735)	KNN	0.9687	0.9687	0.9896	0.9689	0.9687	0.9581	0.9582	1425
	SVM	0.8989	0.8989	0.9668	0.8992	0.8986	0.8649	0.8653	1442
	ANN	0.9267	0.9267	0.9758	0.9264	0.9264	0.9020	0.9021	1475
	RF	0.8854	0.8854	0.9621	0.8865	0.8846	0.8467	0.8476	1504
PSO (1012)	KNN	0.9623	0.9623	0.9875	0.9626	0.9622	0.9496	0.9497	1859
	SVM	0.8968	0.8968	0.9661	0.8964	0.8965	0.8621	0.8621	1881
	ANN	0.9352	0.9352	0.9787	0.9352	0.9352	0.9134	0.9134	2352
	RF	0.8783	0.8783	0.9598	0.8786	0.8770	0.8372	0.8382	1952
LMNN + GWO (829)	KNN	0.9601	0.9601	0.9869	0.9605	0.9602	0.9467	0.9468	2537
	SVM	0.8626	0.8626	0.9549	0.8630	0.8627	0.8164	0.8165	2557
	ANN	0.8883	0.8883	0.9632	0.8878	0.8880	0.8506	0.8507	2569
	RF	0.8085	0.8085	0.9363	0.8051	0.8036	0.7435	0.7455	2627
LMNN + PSO (995)	KNN	0.9552	0.9552	0.9853	0.9554	0.9552	0.9401	0.9401	2772
	SVM	0.8740	0.8740	0.9585	0.8738	0.8730	0.8316	0.8321	2796
	ANN	0.9004	0.9004	0.9671	0.8995	0.8998	0.8668	0.8669	2796
	RF	0.8185	0.8185	0.9399	0.8158	0.8153	0.7571	0.7582	2870

Table 9. Classification results with InceptionV3-based features.

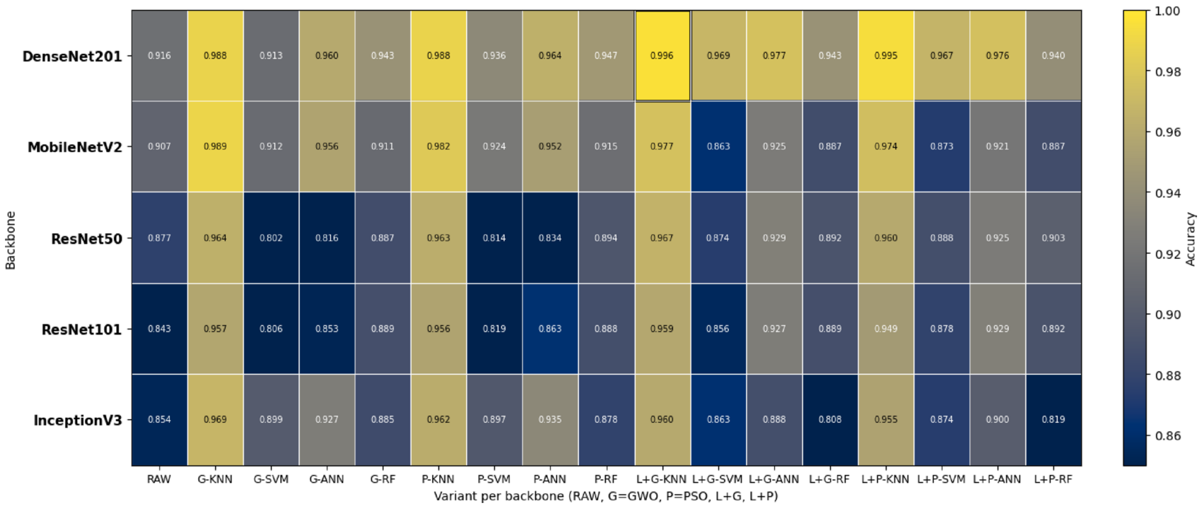


Fig. 4. Comparative analysis of classification accuracy across all models.

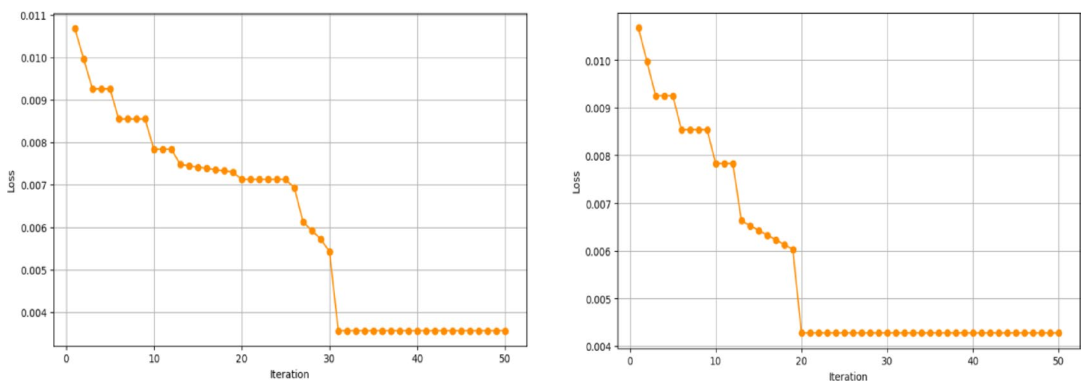


Fig. 5. Loss curves of GWO (left) and PSO (right).

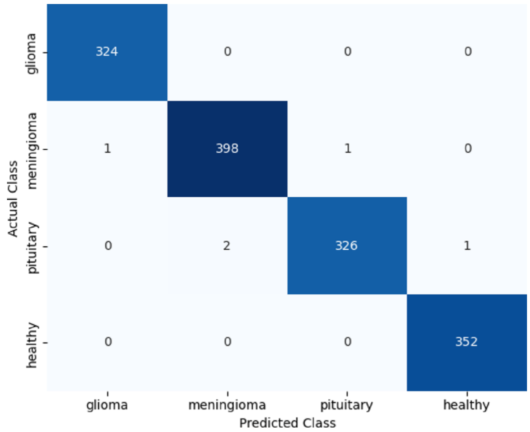


Fig. 6. Confusion matrix of the best-performing pipeline DenseWolf-K.

References	Method/model	Data augmentation	Accuracy (%)
Gómez-Guzmán et al.[28]	Generic CNN	Yes	81.05
Gómez-Guzmán et al.[28]	Inception V3	Yes	97.12
Shilaskar et al.[55]	HOG-XG Boost	Yes	92.02
Rasheed et al.[29]	Image Enhancement + Proposed CNN	Yes	97.84
Celik and Inik[56]	EfficientNetB0-SVM (CNN-optimized ML)	No	97.93
Celik and Inik[56]	CNN-KNN (CNN-optimized ML)	No	97.15
Rahman et al.[57]	LBP feature extraction + CNN	Yes	98.9
Proposed method	DenseWolf-K	Yes	99.64

Table 10. Comparison with state-of-the-art methods for brain tumor classification.

Model	Accuracy	Precision	Sensitivity	F1-Score	Kappa	MCC
DenseNet201	0.867	0.8767	0.867	0.8664	0.8181	0.8212
MobileNetV2	0.789	0.8038	0.789	0.7836	0.7070	0.7137
ResNet50	0.863	0.8704	0.863	0.8622	0.8110	0.8134
ResNet101	0.815	0.8196	0.815	0.8075	0.7434	0.7475
InceptionV3	0.842	0.8421	0.842	0.8366	0.7849	0.7889
Proposed method (DenseWolf-K)	0.933	0.9373	0.933	0.9250	0.9073	0.9109

Table 11. External validation results on the BRISC 2025 dataset.

EfficientNetB0-SVM[56] and LBP + CNN[57], achieve accuracies close to 98–99%. In comparison, the proposed DenseWolf-K method attains 99.64%, surpassing all prior works. This superior performance confirms the benefit of integrating metric learning and swarm-based feature selection into the classification pipeline.

External validation is a critical step in evaluating the generalization ability of machine learning models, as it demonstrates how well a trained system performs on previously unseen, independent datasets. Unlike internal validation, which may still carry dataset-specific biases, external validation provides stronger evidence of robustness and real-world applicability. For this purpose, the BRISC 2025[58,59] dataset was employed. The trained models were tested on this independent dataset, and the results are presented in Table 11. Accordingly, while baseline CNN architectures achieved accuracies in the range of 78.9–86.7%, the proposed DenseWolf-K pipline substantially outperformed all baselines with an accuracy of 93.3%, an F1 score of 0.925, and an MCC of 0.9109. These findings highlight the robustness and generalization capacity of the proposed approach across independent data sources.

In summary, the results show that while all CNN backbones provided solid baselines, DenseNet201 consistently achieved the best performance. When combined with LMNN, swarm-based feature selection, and KNN, the proposed DenseWolf-K pipeline reached 99.64% accuracy with compact feature subsets, confirming both efficiency and precision. GWO generally outperformed PSO by delivering higher accuracy with fewer features and shorter computation times. External validation on the BRISC 2025 dataset further demonstrated the robustness and generalizability of the framework. Overall, DenseWolf-K surpassed state-of-the-art approaches while offering feature- and image-level interpretability, highlighting its potential as a reliable and clinically relevant solution for MRI-based brain tumor classification.

Explainability analysis

This section presents the explainability analyses conducted to enhance the transparency of the proposed framework, encompassing both feature-based and image-based perspectives. By integrating local and global analyses, the pipeline offers dual-level interpretability that addresses the “black box” criticism of deep learning models, thereby improving its clinical relevance and trustworthiness.

Feature-based explanations

Deep learning models are often criticized for operating as opaque “black boxes,” where their internal decision-making processes remain inaccessible to end users. In clinical applications, this lack of transparency constitutes a major barrier to adoption, as healthcare professionals must understand why a model produces a particular prediction before it can be trusted in diagnostic workflows[60,61]. To address this limitation, the proposed pipeline integrates explainability analyses at both the local and global levels.

Figure 7a illustrates a local explanation of a misclassified case, where the true label was *glioma* but the model predicted *meningioma*. The bar plot displays the normalized contribution of latent CNN-derived features to this specific decision. Among them, features such as *f933* and *f1563* exerted the strongest influence, ultimately driving the model toward an incorrect classification. Such patient-level insights are valuable as they reveal which latent attributes may mislead the model in challenging cases, thereby exposing potential vulnerabilities in the learned feature space and guiding error analysis.

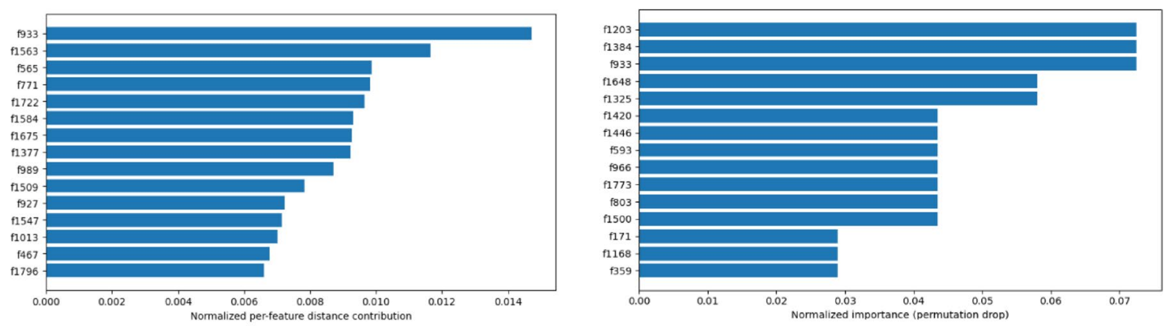
Figure 7b presents the results of a global feature importance analysis, obtained using a permutation-based ranking in the LMNN feature space. Features such as f1203, f1384, and f933 consistently ranked among the most influential across the entire dataset. By quantifying the relative contribution of features to model accuracy, this analysis identifies the deep features that are most critical for distinguishing between tumor types. Global importance plots therefore complement patient-level explanations by providing broader insight into the decision-making logic of the pipeline.

Taken together, these results demonstrate that the proposed framework does not solely aim for high classification accuracy but also incorporates interpretable mechanisms that reveal its inner workings. By combining local and global perspectives, the model provides meaningful insights into both individual predictions and general decision patterns, thereby strengthening its potential utility as a clinically accountable AI tool.

Image-based explanations

In addition to feature-level interpretability, Occlusion Sensitivity Analysis[62,63]—a perturbation-based explainable-AI (XAI) method—was applied to the DenseWolf-K pipeline to obtain image-level explanations Each MRI was tiled with square patches; at each step one patch was masked and the pipeline was re-evaluated. Regional importance was quantified as the normalized drop in the KNN decision margin in LMNN space, $\Delta\,(d_2 - d_1)$ where d_1 and d_2 are the distances to the first and second nearest neighbors, respectively. The resulting heatmaps highlight regions most critical to the prediction, with warmer colors indicating occlusions that cause larger margin reductions.

Figure 8 presents representative examples from each class (glioma, healthy, meningioma, pituitary). In the glioma case, salient regions co-localize with intraparenchymal tumor tissue; in healthy scans, the maps emphasize normal cortical/meningeal boundaries and brain–skull interfaces, consistent with robust negative decisions. For meningioma, hot spots appear along dural/extra-axial attachments, reflecting class-specific morphology, while in the pituitary case the sellar–parasellar region is prominently highlighted. These occlusion maps provide radiologists with intuitive visual evidence of the anatomical drivers of the classification, thereby demonstrating the contribution of an established XAI technique and enhancing the clinical trustworthiness of the proposed framework.



(a) **(b)**

Fig. 7. Local and global feature explanations. (**a**) Local explanation of a misclassified instance (true glioma, predicted meningioma). (**b**) Global feature importance in the LMNN features space.

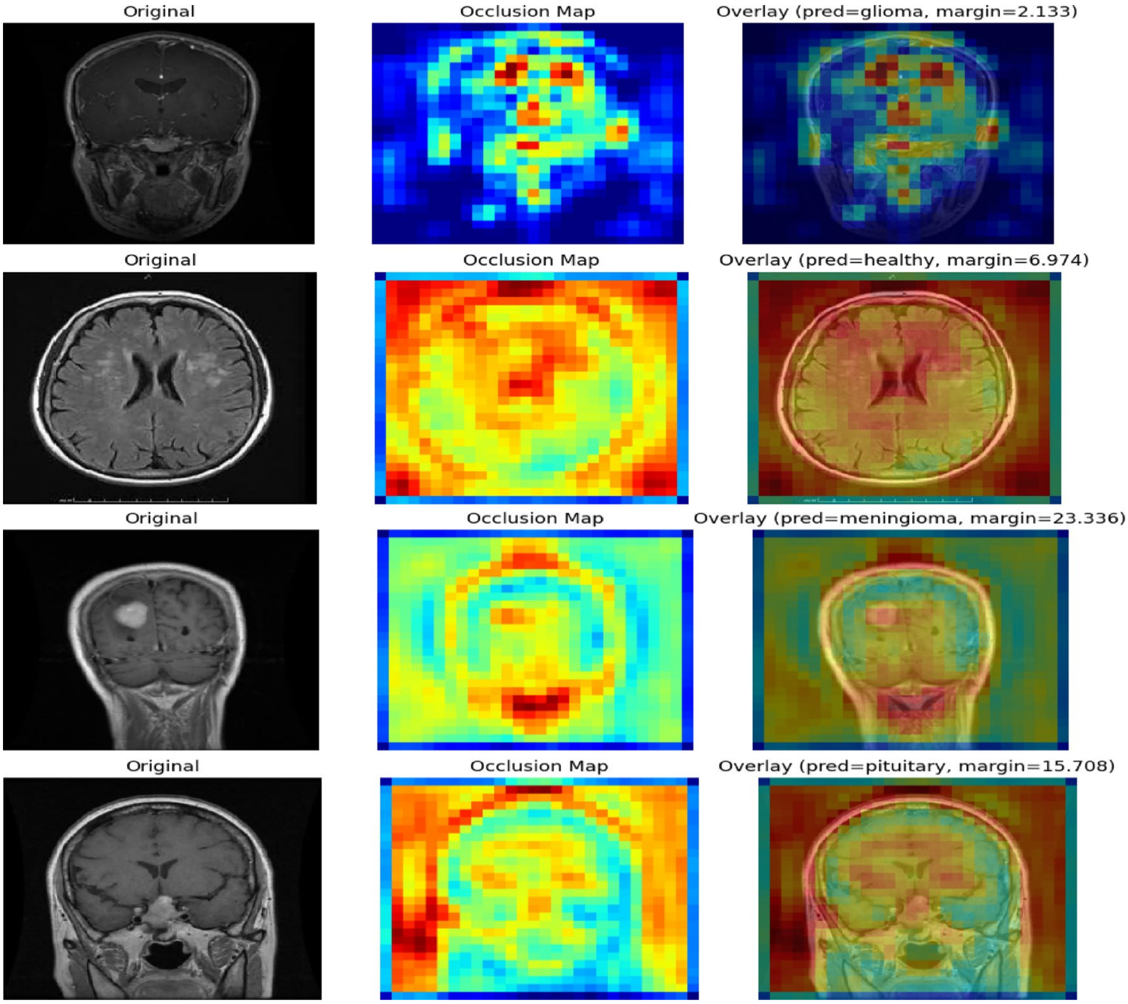


Fig. 8. Occlusion sensitivity maps for representative cases from each class.

In summary, the DenseWolf-K pipeline combines high predictive performance with feature- and image-level XAI, enhancing its potential for clinical decision support.

Conclusion

This study proposed a novel hybrid framework that combines deep feature extraction, metric learning, and swarm-based feature selection to improve brain tumor classification from MRI images. Five pretrained CNNs—DenseNet201, MobileNetV2, ResNet50, ResNet101, and InceptionV3—were systematically evaluated through five-fold cross-validation, with DenseNet201 achieving the strongest baseline performance. The initial findings indicated that while all CNNs demonstrated promising accuracy, not all extracted features contributed equally to classification, motivating the development of a feature selection and refinement strategy.

To address redundancy and improve separability, Particle Swarm Optimization (PSO) and Grey Wolf Optimizer (GWO) were applied both before and after LMNN transformation. This dual-stage design reduced dimensionality while enhancing class discriminability. The DenseNet201–LMNN–GWO–KNN configuration, termed DenseWolf-K, achieved the highest performance with 99.64% accuracy, establishing it as the optimal implementation of the framework. Between the two optimizers, GWO consistently outperformed PSO by selecting fewer features and requiring shorter computation times. LMNN's contribution was shown to be architecture-dependent, yielding notable improvements for DenseNet201 and ResNet50, while having more limited effects on MobileNetV2 and InceptionV3.

The framework demonstrated robustness across classifiers (KNN, SVM, ANN, RF) and metrics, outperformed or matched recent state-of-the-art methods, and generalized well to an external dataset. Explainability was ensured through feature-level ranking of GWO-selected features in the LMNN space and image-level occlusion sensitivity heatmaps, thereby supporting clinical interpretability.

In conclusion, the dual-stage integration of LMNN with swarm-based optimization constitutes a unique methodological contribution, as no prior brain tumor classification study has systematically examined optimization both before and after metric transformation. These advances highlight DenseWolf-K as a reliable, interpretable, and scalable candidate for clinical imaging workflows.

Future directions include validation on larger, multi-center, and multimodal datasets to further strengthen generalizability, as well as the creation and use of federated or harmonized datasets to overcome data scarcity and domain shift issues. In addition, advancing explainable AI from 2D occlusion maps toward volumetric or temporal explanations aligned with lesion ROIs and clinician studies will enhance the interpretability of the framework. Another promising line of research is reducing computational complexity through lightweight CNNs and parallelized implementations to enable real-time deployment in clinical workflows. Finally, extending the proposed pipeline to other imaging tasks, such as lung or breast cancer, will demonstrate its broader clinical value and applicability.

Data availability

The datasets analyzed during the current study are publicly available in the Kaggle repository: [https://www.kaggle.com/datasets/rm1000/brain-tumor-mri-scans/data] and [https://www.kaggle.com/datasets/briscdataset/brisc2025/data].

Received: 5 May 2025; Accepted: 9 October 2025
Published online: 28 October 2025

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Author contributions
A.Y. conceived the study, performed data preprocessing, implemented the models, conducted all experiments, analyzed the results, and wrote the manuscript. A.Y. also prepared all figures and approved the final version of the manuscript.

Funding
The open access fee of this article was supported by the Selçuk University Scientific Research Projects (BAP) Coordination Unit under Project No. 25601063.

Declarations

Competing interests
The author declares no competing interests.

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